

INTRO. TO FDAs

THREE KEY STEPS IN SOLV OF ANY PROBLEM USING FDAs

- 1) DOMAIN: DISCRETIZATION OF CONTINUOUS INDEPENDENT VARs
- 2) DISCRETIZATION OF EQUATIONS: REPLACE DERIVATIVES WITH FDAs
- 3) SOLUTION: SOLVE RESULTING ALGEBRAIC EQNS FOR APPROXIMATE SOLN VALUES, DEFINED ON DISCRETE DOMAIN (HERE, ONCE AN INITIAL CONDITION, $x(0)$, IS SPECIFIED)

2. MATHEMATICAL PRELIMINARIES

2.1 "BIG-O" NOTATION

- DEFN: $f(h)$ is $O(h^r)$ IF AND ONLY IF THERE ARE 2 POSITIVE REAL NUMBERS, M, h_0 SUCH THAT

$$|f(h)| \leq M |h^r| \text{ FOR ALL } h < h_0$$

THINK OF h AS A SMALL QUANTITY SO $h^{r+1} < h^r$

2.2 TAYLOR SERIES

- TAYLOR SERIES (TS) EXPANSION AND FDAs

— CAN BE USED TO DERIVE FDAs

— CAN BE USED TO ESTABLISH ACCURACY OF FDA: *

- WILL USUALLY USE TS TO EXPAND EXPRESSIONS SUCH AS $f(x+h)$ ABOUT x . TAYLOR'S THM TAKES THE FORM

$$f(x+h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} h^n = \sum_{n=0}^{\infty} h^n \frac{f^{(n)}(x)}{n!} \quad (\text{TS})$$

h IS THE EXPANSION PARAMETER

$f^{(n)}(x) \equiv n$ -TH DERIVATIVE OF f EVALUATED AT x

EXPLICITLY:

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{6} f'''(x)$$

$$+ \frac{h^4}{24} f^{(4)}(x) + O(h^5)$$

3. DISCRETIZATION: STEP 1 - F.D. GRIDS (MESHES, LATTICES)

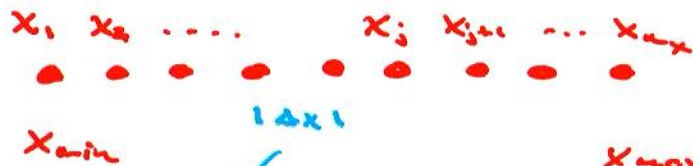
- START WITH FENS IN ONE DIMENSION; I.E. ONE INDEPENDENT VBL. (TYPICALLY x OR t)
- SOLN OF SOME DIFFERENTIAL EQN ON SOME FINITE DOMAIN

• EXAMPLES

$$- x_{min} \leq x \leq x_{max}$$

$$- 0 \leq t \leq t_{max}$$

• UNIFORM FINITE DIFFERENCE GRID x_j



CONSTANT, SAME Δx BETWEEN ANY TWO ADJACENT GRID POINTS

- NUMBER OF GRID PTS; N_x

- MESH SPACING

$$\Delta x = \frac{x_{max} - x_{min}}{N_x - 1}$$

- GRID POINTS

$$x_j = x_{min} + (j-1)\Delta x, \quad j=1, 2, \dots, N_x$$

- GRID FUNCTION NOTATION; f_j

$$f_j \equiv f(x_j)$$

- THEN, HAVE E.C.

$$f_{j+1} = f(x_j + \Delta x)$$

$$f_{j-1} = f(x_j - \Delta x)$$

$$f_{j+2} = f(x_j + 2\Delta x)$$

4. DISCRETIZATION: STEP 2 - DERIVATION OF FDs

- CONSIDER 4 FDs
 - 3 FOR FIRST DERIVATIVE
 - 1 FOR SECOND "

• FIRST, WRITE DOWN AND DEMONSTRATE LEVEL OF ACCURACY, THEN LOOK AT ONE METHOD FOR DERIVATION

4.1 FIRST ORDER FORWARD FDA FOR FIRST DERIV. $f'(x)$

$$f'(x) \approx \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

- "FORWARD DIFFERENCE" SINCE WE USE VALUE "FORWARD" OF x , I.E. $x+\Delta x$ TO COMPUTE THE APPROXIMATION

$$f'_j \equiv f'(x_j) \approx \frac{f_{j+1} - f_j}{\Delta x}$$

→ DERIVATES IS REPLACED BY ; GET 1st FORMULA

$$\boxed{f'(x_j) \rightarrow \frac{f_{j+1} - f_j}{\Delta x}} \quad (1)$$

• ACCURACY?

- USE TAYLOR SERIES: $h \rightarrow \Delta x$ IN EQN (1)

$$f(x+\Delta x) = f(x) + \Delta x f'(x) + \frac{\Delta x^2}{2} f''(x) + \frac{\Delta x^3}{6} f'''(x) + O(\Delta x^4)$$

$$\frac{f(x+\Delta x) - f(x)}{\Delta x} = \underbrace{f'(x)}_{\checkmark} + \frac{1}{2} \Delta x f''(x) + \frac{1}{6} \Delta x^2 f'''(x) + O(\Delta x^3)$$

INCREASING ORDER ERROR
TERM OF Δx
I.E. FIRST ORDER

$$= f'(x) + \frac{1}{2} \Delta x f''(x) + O(\Delta x^2)$$

$$= f'(x) + O(\Delta x)$$

- ERROR TERM $O(\Delta x)$ MEANS WE HAVE FIRST ORDER APPROX FOR $f'(x)$ AT x
- I.E. AS $\Delta x \rightarrow 0$ ERROR IN APPROXIMATION WILL TEND TO DECREASE LINEARLY IN Δx E.G. $\Delta x \rightarrow \Delta x/2$ ERROR $\rightarrow$ ERROR $/2$

4.2 FIRST ORDER BACKWARDS FDA FOR $f'(x)$

- NOW "BACKWARDS" DIFFERENCE

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

- GRID FCN NOTATION

$$f'_j \equiv f'(x_j) \approx \frac{f_j - f_{j-1}}{\Delta x}$$

- SECOND FORMULA FOR $f'(x)$

$$\boxed{f'(x_j) \rightarrow \frac{f_j - f_{j-1}}{\Delta x}} \quad (2)$$

- ACCURACY?

$$f(x - \Delta x) = f(x) - \Delta x f'(x) + \frac{\Delta x^2}{2} f''(x) - \frac{\Delta x^3}{6} f'''(x) + O(\Delta x^4)$$

- FROM (2) HAVE

$$\begin{aligned} \frac{f(x) - f(x - \Delta x)}{\Delta x} &= \cancel{f'(x)} - \frac{1}{2} \Delta x f''(x) + O(\Delta x^2) \\ &= f'(x) + O(\Delta x) \end{aligned}$$

- ALSO FIRST ORDER

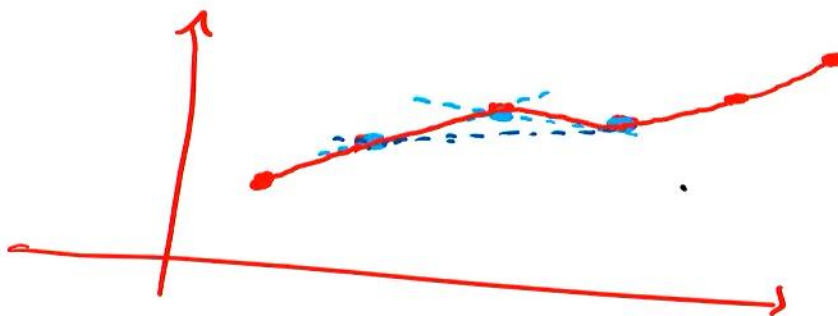
4.3 SECOND ORDER CENTRED FOR $f'(x)$

• INTUITIVELY, GET HIGHER ORDER DERIVATIVE BY AVERAGING TWO PREVIOUS FORMULAE

• TRY

$$\frac{1}{2} \left(\frac{f(x+\Delta x) - f(x)}{\Delta x} + \frac{f(x) - f(x-\Delta x)}{\Delta x} \right) \\ = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}$$

$$\left| f'(x_j) - \frac{f_{j+1} - f_{j-1}}{2\Delta x} \right| \quad (3)$$



• Accuracy?

• TAYLOR SERIES

$$f(x+\Delta x) = f(x) + \Delta x f'(x) + \frac{\Delta x^2}{2} f''(x) + \frac{\Delta x^3}{6} f'''(x) + O(\Delta x^4) \\ f(x-\Delta x) = f(x) - \Delta x f'(x) + \frac{\Delta x^2}{2} f''(x) - \frac{\Delta x^3}{6} f'''(x) + O(\Delta x^4)$$

• FROM (3) HAVE

$$\frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} = f'(x) + \frac{1}{6} \Delta x^2 f'''(x) + O(\Delta x^4) \\ = f'(x) + O(\Delta x^2)$$

SECOND ORDER
APPROXIMATION

• AS $\Delta x \rightarrow 0$ ERROR DECREASES QUADRATICALLY
IN Δx . $\Delta x \rightarrow \Delta x/2$; ERROR $\rightarrow$ ERROR/4

- "CENTRED" BECAUSE STRUCTURE OF FORMULA IS SYMMETRIC ABOUT POINT OF APPROXIMATION